Unit III: WIRELESS LANS

Infra red vs radio communication

- Uses infrared light waves (short-range, line-of-sight). Common in remote controls, short-range data links.

Advantages:

- Cheap and simple.
- No licensing required.

Disadvantages:

- Blocked by obstacles.
- Not suitable for mobility across rooms.

2. Radio Waves: Advantages:

- Wide coverage.
- Supports mobility.

Disadvantages:

- Prone to **interference and noise**.
- Requires spectrum regulation.

Feature	Infrared	Radio
Medium	Light waves	Radio waves
Range	Short	Long
Line-of-sight	Required	Not required
Penetration	Cannot pass walls	Can pass walls
Interference	Low	High
Security	High	Moderate

Infrastructure vs Ad-hoc Network

Infrastructure Network: a wireless network where all devices communicate through a central **Access Point (AP)** that connects to other networks.

```
     Internet
        |
  +-----------+
  | Access    |
  | Point(AP) |
  +-----------+
   /  |  \
  STA STA STA
```

Features:

- Devices communicate through AP.
- AP connects to a wired network.

Pros:

- Better management.
- Scalable.

Cons:

- Depends on AP (single point of failure).

Ad-hoc Network

- No central controller.
- Devices communicate directly.

```
STA ------ STA
 | \    / |
 |  \  /  |
 |   \/   |
 |   /\   |
 |  /  \  |
STA ------ STA
```

(No Access Point)

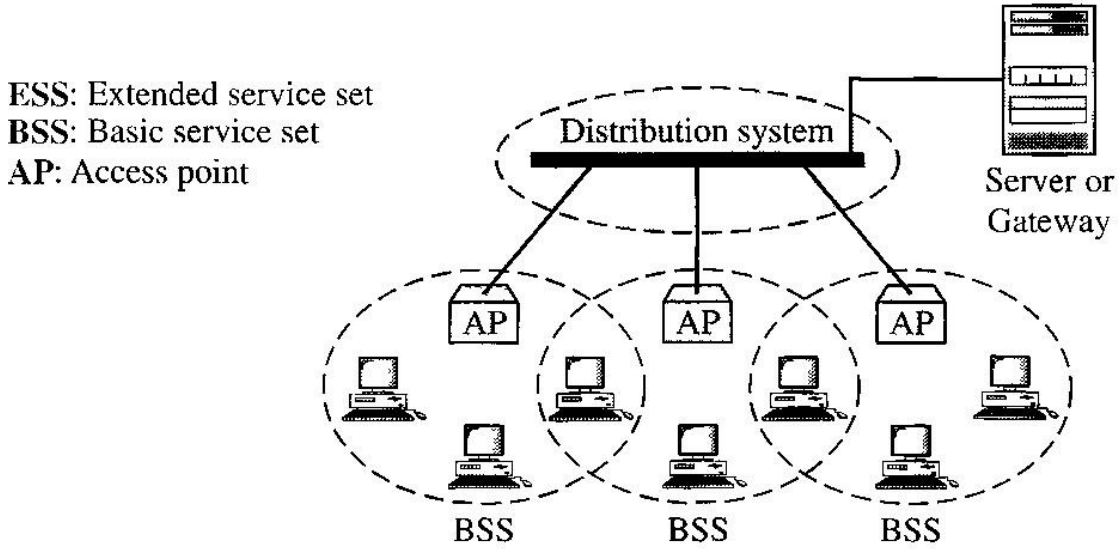
Features:
- Self-configuring.
- Temporary networks.

Pros:
- Flexible.
- No infrastructure needed.

Cons:
- Limited scalability.
- Complex routing.

3. IEEE 802.11 (Wi-Fi): *Institute of Electrical and Electronics Engineers*

System Architecture Defined by IEEE 802.11



Components:
- Station (STA): Wireless device.

- Access Point (AP): Central node.
- Basic Service Set (BSS): Group of stations.
- Extended Service Set (ESS): Multiple BSS interconnected.

Hidden Node Problem in WLAN

Two nodes cannot hear each other but communicate with same AP → collision occurs.

Node A ---> AP <--- Node B (A
and B cannot hear each other)

Solution:

- RTS/CTS mechanism: It is a **handshaking process** before actual data transmission: ● **RTS** → Sender asks permission to send
- **CTS** → Receiver grants permission

Working Steps

1. Sender → RTS → Receiver
2. Receiver → CTS → Sender
3. Sender → DATA → Receiver
4. Receiver → ACK → Sender

Impact on WLAN

1. Collisions at the receiver

- A and C may transmit at the same time.
- Since they don't detect each other, both assume the channel is free.
- Their signals collide at the Access Point → **packet loss**.

2. Reduced throughput

- Collisions require retransmissions.
- More retransmissions = **lower effective data rate**.

3. Increased latency

- Backoff algorithms (like CSMA/CA) delay retransmissions.
- Results in **longer delays**, especially under heavy load.

4. **Unfair bandwidth usage**

- Some nodes may experience more collisions than others.

- Leads to **uneven performance** across devices.

5. Energy inefficiency

- Repeated retransmissions consume more battery power in mobile devices.

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2. Exposed Terminal Problem occurs when a node refrains from transmitting because it senses another transmission—even though its transmission wouldn't actually cause interference.

Example:

- Node B → Node A
- Node C → Node D
- B and C can hear each other, but A and D are far apart.
- C unnecessarily stays silent when B transmits.

Impact on WLAN

1. Underutilization of bandwidth

- Nodes avoid transmitting even when it's safe.
- Leads to **wasted channel capacity**.

2. Reduced network throughput

- Fewer simultaneous transmissions than possible.
- Overall **network efficiency drops**.

3. Increased delays

- Nodes wait unnecessarily → **longer transmission times**.

4. Lower spatial reuse

- WLANs depend on reusing spectrum across space.
- Exposed terminal problem limits this → **reduced scalability**.

Mitigation Techniques

- RTS/CTS (in some cases helps distinguish safe transmissions)
- Advanced MAC protocols (e.g., adaptive carrier sensing)
- Directional antennas or smarter sensing algorithms

3. Comparative Impact

Aspect	Hidden Terminal	Exposed Terminal
Main issue	Collisions	Unnecessary waiting
Effect on throughput	Decreases (due to retransmissions)	Decreases (due to idle time)
Channel usage	Inefficient (overlapping signals)	Inefficient (unused capacity)
Delay	High (retries)	Moderate (waiting)
Energy use	High	Moderate

```
Internet
   |
 Router
   |
Access Point
 /  |  \
Laptop Phone Tablet
```

Protocol Architecture 802.11
follows layered design:

```
+-----------------------------+
|    Application Layer        |
+-----------------------------+
|    Transport (TCP/UDP)      |
+-----------------------------+
|    Network (IP)             |
+-----------------------------+
|    LLC (Logical Link)       |
```

```
+-----------------------------+
|    MAC Layer (802.11)      |
+-----------------------------+
|    Physical Layer          |
+-----------------------------+
```

Physical Layer

Functions:

- Signal transmission and reception.
- Modulation techniques (DSSS, OFDM).

Key features:

- Frequency bands: 2.4 GHz, 5 GHz.
- Data rates depend on standard (a/b/g/n/ac/ax).

Direct Sequence Spread Spectrum DSSS is a spread spectrum technique where the original data signal is multiplied by a **high-rate pseudo-random bit sequence (chip sequence)** to spread it over a wider bandwidth. Ex: sending duplicates in error prone channel.

OFDM (Orthogonal Frequency Division Multiplexing) is a multicarrier modulation technique where data is split into multiple **parallel subcarriers**, each transmitted at a lower data rate. Ex: sending fragments in narrow channel

Feature	DSSS	OFDM
Technique Type	Spread spectrum	Multicarrier modulation
Bandwidth Use	Spread using code	Split into many subcarriers
Interference Handling	Strong against narrowband interference	Strong against multipath fading
Complexity	Lower	Higher
Typical Use	Older wireless systems, GPS	Modern high-speed communication

MAC Layer (Medium Access Control)
Key Protocol: CSMA/CA
(Carrier Sense Multiple Access with Collision Avoidance)

Working:

1. Listen before transmit.
2. If busy → wait.
3. Random backoff time.
4. Transmit when free.

```
  Channel Idle?
      |
     Yes
      |
  Wait DIFS
      |
  Random Backoff
      |
  Channel Free?
    /   \
   No     Yes
  |      |
 Wait      Transmit Frame
          |
       ACK Received?
       /      \
      Yes      No
      |        |
```

Success Retransmit

DIFS = Distributed Interframe Space
It is a **time interval** used in the MAC layer of IEEE 802.11 to control **when a station is allowed to transmit data**.

Why CA (not CD)?

- Collision detection is difficult in wireless.

MAC Management
Handles:

- Association / Reassociation
- Authentication
- Synchronization
- Power management **Functions:**

Medium access (CSMA/CA)
Fragmentation and reassembly
Error control (ACKs)
Frame addressing
Security (encryption/authentication)

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4. Bluetooth: a short-range PANs wireless communication technology used for exchanging data between devices over 2.4 GHz, with layered architecture (Radio → Baseband → LMP → L2CAP),

User Scenarios

- Wireless headset
- File transfer between phones
- IoT device communication
- Smart home devices

Architecture
Basic unit: Piconet

- One master + up to 7 slaves.
- Multiple piconets → Scatternet

```
        Headset
           |
Laptop --- Smartphone --- Smartwatch
           |
        Tablet
```

(Smartphone = Master)
Phone acts as:

- **Master in one piconet**
- **Slave in another**

→ Forms a **scatternet**

```
+-----------------------------+
|    Applications             |
+-----------------------------+
|    L2CAP                    |
+-----------------------------+
|    Link Manager Protocol    |
|       (LMP)                 |
+-----------------------------+
|    Baseband Layer           |
+-----------------------------+
|    Radio Layer              |
+-----------------------------+
```

Radio Layer

- Frequency: 2.4 GHz ISM band
- Uses Frequency Hopping Spread Spectrum (FHSS) Features:
- 1600 hops/sec

- Reduces interference

Baseband Layer ● Handles physical links.

- Two link types:
  - SCO (Synchronous Conn. Oriented): voice
  - ACL (Asynchronous Conn. Less): data

Link Manager Protocol (LMP) ● Controls:

- Link setup
- Authentication
- Encryption
- Power control

L2CAP (Logical Link Control and Adaptation Protocol) Functions:

- Multiplexing multiple connections.
- Segmentation and reassembly.
- QoS support.

Feature	Wi-Fi (802.11)	Bluetooth
Range	Long	Short
Speed	High	Moderate
Power	High	Low
Use	Internet access	Device connectivity